

Cedric PIZINA

Statistics and Data Science | Climate Dynamics

📍 Central, SC

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Profile

Graduate student in Statistics and Data Science at Clemson University with an interdisciplinary background in climate physics, hydraulic engineering and mathematical sciences. Passionate about climate-data analysis, numerical and statistical modelling. Recognized as a top-performing student and valued for adaptability, rapid learning the ability and unique background.

Education

Clemson University **Aug. 2026 – Present**
Clemson, SC

PhD Program in Statistics and Data Science

International Centre for Theoretical Physics (ICTP) **Sep. 2025 – Aug. 2026**
Trieste, Italy

Pre-PhD Diploma in Physics of Climate and Earth System Physics

GPA: 4.0

AIMS South Africa | Stellenbosch University **Sep. 2024 – Jul. 2025**
Cape Town, South Africa

Structured Master's in Mathematical Sciences

GPA: 4.0

Ecole Supérieure Polytechnique d'Antananarivo | University of Antananarivo **Jan. 2022 – Aug. 2024**
Antananarivo, Madagascar

Master of Engineering in Hydraulics and Development

GPA: 3.85

Ecole Supérieure Polytechnique d'Antananarivo | University of Antananarivo **Jul. 2018 – Aug. 2021**
Antananarivo, Madagascar

Bachelor of Engineering in Hydraulics and Fluid Mechanics

GPA: 3.85

Research Experience

Postgraduate Diploma Thesis **Sep. 2025 – Aug. 2026**

Uncovering the failure of the early-winter IOD–NAO teleconnection in SEAS5 and its sensitivity to different atmospheric mean states

- Analyzed climate-model output using partial regression approach to identify the causes of teleconnection failure;
- Conducted mean states sensitivity experiments using dry Linear Baroclinic Model.

MSc Project **Mar. 2025 – Jun. 2025**

Monitoring Southern African droughts under El Niño–Southern Oscillation (ENSO) and Indian Ocean Dipole (IOD) impacts

- Analyzed the joint impact of ENSO and IOD phases on Southern Africa's climate during 1980–2023.
- Assessed future trends in precipitation, temperature, and soil moisture through 2100 using CMIP6 models.
- Forecasted future mean trends in the Southern African Drought Index using machine learning.

MEng Project **Jul. 2023 – May 2024**

Diagnostic of the drinking-water network of Sainte Marie, Madagascar

- Led site data collection and diagnostic analysis to identify operational problems and their causes;
- Developed an optimized EPANET model for system diagnostic and optimization;
- Proposed improvements to network efficiency.

Teaching and Professional Experience

Clemson University

Aug. 2026 – Present
Clemson, SC

Lab Instructor, Introduction to Statistics (STAT 2300)

- Conduct weekly laboratory sessions using JMP and grade laboratory assignments.

GRET Madagascar

Jul. 2023 – May 2024
Antananarivo, Madagascar

Junior Hydraulic Engineer

- Contributed to the design of drinking-water systems and watershed protection plans for Ambatofotsy and Vinaninony.
- Conducted flow, pressure, and water-quality measurements at production sites in Andramasina and Sainte Marie.
- Contributed to field data collection by developing surveys and coordinating on-site quality controls.

Selected Coursework

Geophysical Fluid Dynamics	Atmospheric Physics	Numerical Methods	Earth System Modeling
Ocean Dynamics	Biogeochemical Cycles	Applied Statistical Models	Statistical Inference

Awards and Leadership

Mastercard Foundation Scholarship | AIMS South Africa

Sep. 2024 – Jul. 2025

Mastercard Foundation Scholar Network Member

Awarded a fully funded scholarship to complete a master's program at AIMS and develop leadership skills through community service.

Academic Class Prize

Jul. 2018 – Jul. 2025

Top achiever in undergraduate and graduate studies

Recognized as a top achiever at the Ecole Supérieure Polytechnique d'Antananarivo and received an academic excellence award at AIMS South Africa.

Community Engagement and Workshops

Muizenberg Library | AIMS South Africa

Jan. 2025 – Jul. 2025
Cape Town, South Africa

Mathematics and Physics Tutor

Tutored middle- and high-school students in mathematics and physics.

AIMS House of Science | African Gong

Apr. 2025
Cape Town, South Africa

Africa Scientifique 2025

Attended a three-day workshop on scientific communication, discovery, and capacity building.

University of Cape Town | AIMS South Africa

Sep. 2024
Cape Town, South Africa

Design Thinking

Completed a two-day workshop focused on design thinking and teamwork skills.

Technical Skills

Programming	Python, Fortran, SageMath, R, $\text{\LaTeX}$
Engineering and GIS	AutoCAD, EPANET, QGIS, ArcGIS, HEC-HMS, HEC-RAS
Office	Microsoft Word, Excel, PowerPoint, and Outlook
Languages	Malagasy (native), French (C1), English (C1)

References

Dr. Fred Kucharski

International Centre for Theoretical Physics, Italy
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Dr. Riccardo Farneti

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